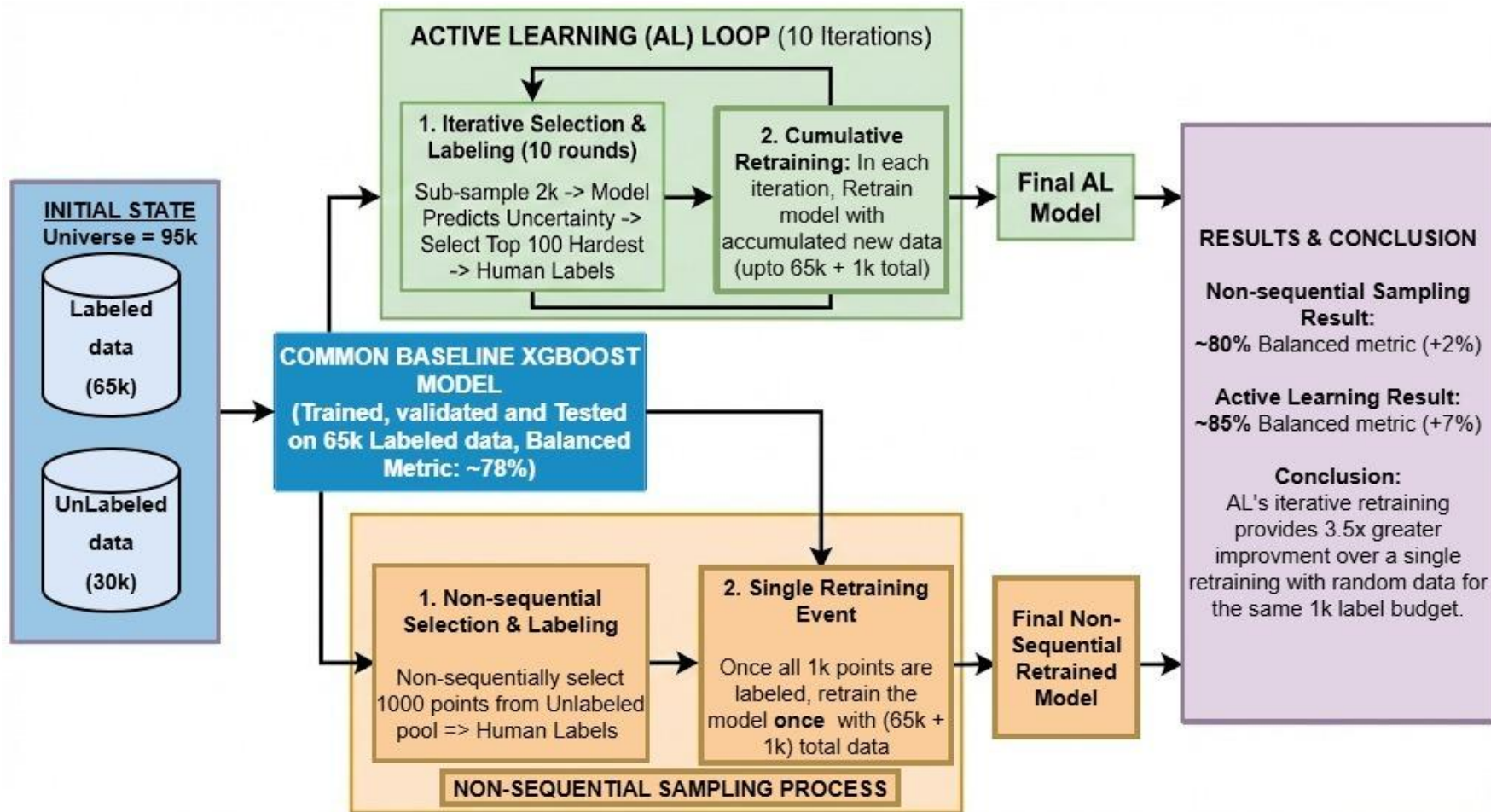


Insurance Fraud Detection using AL algorithm

Presented by:
Prof. Nandan Sudarsanam
Aakash Sarraf
Sujaal Burad

Overall framework



Core Objective

- **Maximize ROI on Labeling:** Comparing how to best spend a budget of **1,000 new human labels** to improve an XGBoost baseline.
- Testing if *what* we label matters as much as *how many* we label.
- Shift to **financial loss** (including costs for TP, TN, FP, FN cases) as metric from balanced accuracy that was used for baseline.

AL Gain

- **Uncertainty Sampling:** Instead of random selection, the model "asks" for labels on the **top 100 most difficult/uncertain** cases in each of the 10 rounds.
- **Iterative Evolution:** The model is retrained 10 times, allowing it to "learn as it goes" and refine its uncertainty with every batch.
- Intelligence-driven data selection significantly outperforms brute-force data labeling.

We compare the model trained on random 1000 points to that on selected 100*10 points via AL to establish a better learning ability and results. Hence we need 10 to-and-fro iterations for true labels with the bank for the requested 100 + random 100 points in each iteration.

Weighted cost

- **COST_TP = 100** $\Leftrightarrow$ model says fraud (1), actually fraud (1) $\Rightarrow$ investigation cost
- **COST_FP = 150** $\Leftrightarrow$ model says fraud (1), actually not fraud (0) $\Rightarrow$ false alarm cost
+ reputation damage
- **COST_TN = 0** $\Leftrightarrow$ model says not fraud (0), actually not fraud (0) $\Rightarrow$ no cost
- **COST_FN = 0.9 x sum_insured** $\Leftrightarrow$ model says not fraud (0), actually fraud (1) $\Rightarrow$ missed fraud = pay out 90% of claim

See References B9, B10, B11.

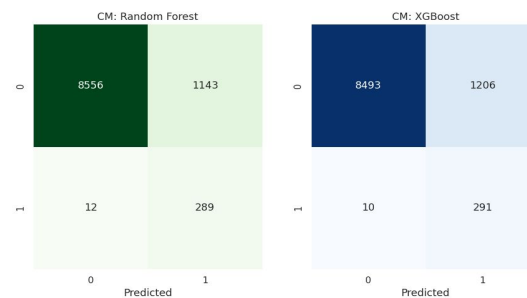
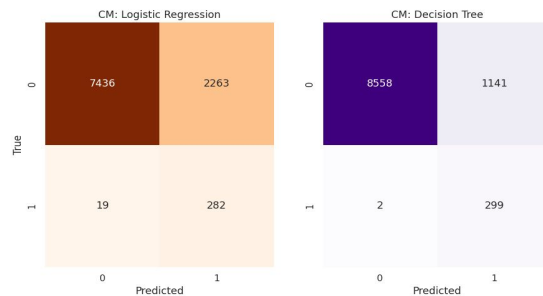
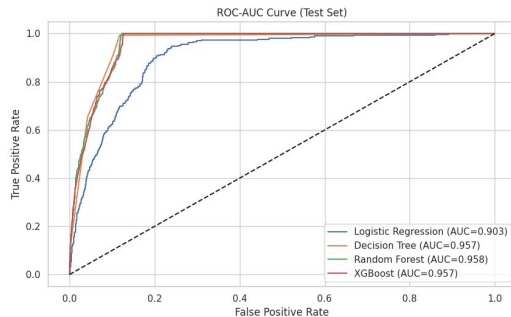
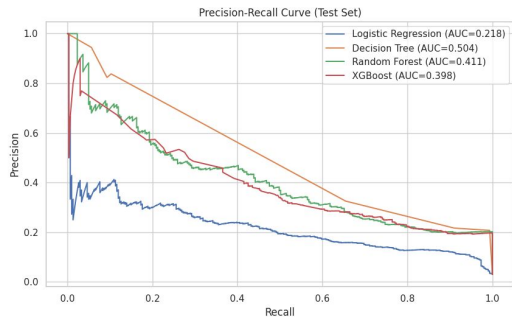
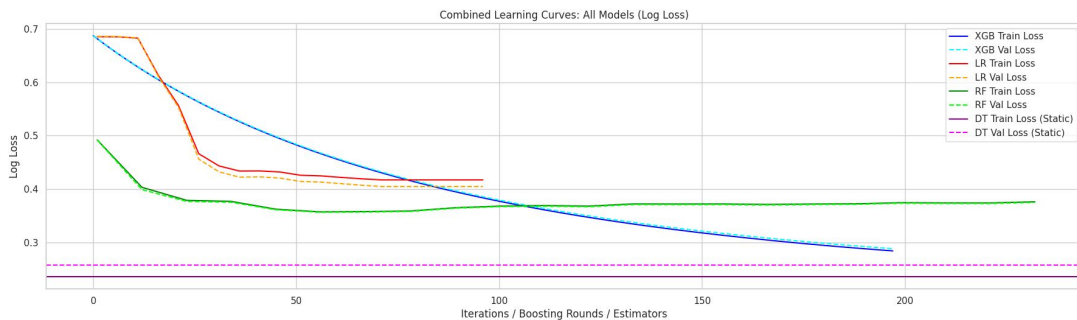
Code Flow

- 1] Grid search (Using optuna) – To find the optimal common baseline XGBoost model.
- 2] Active Learning Loop – Run the code block 10 times (Needs manual input of fraud labels for the uncertain points)
- 3] Non-sequential branch – Run the code block 1 time (Needs manual input of randomly selected 1k data points)

Results

Case 1: Train (45k), val (10k) and test (10k)
from labeled set only

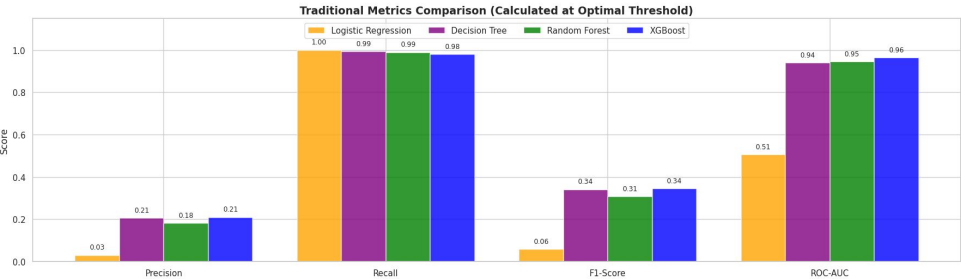
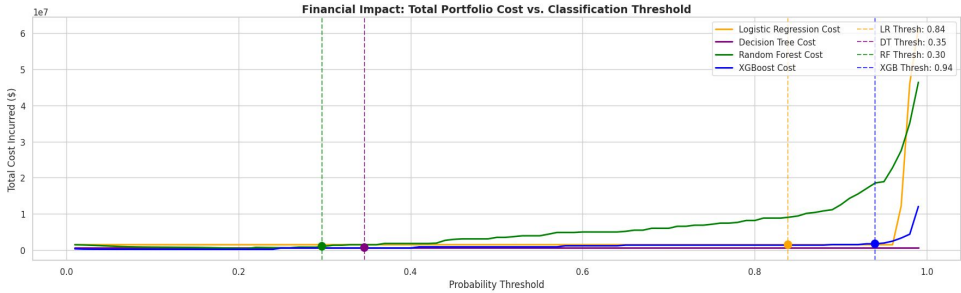
	Model	ROC-AUC	F1-Score	Precision	Recall
0	Logistic Regression	0.903196	0.198173	0.110806	0.936877
1	Decision Tree	0.956934	0.343481	0.207639	0.993355
2	Random Forest	0.957858	0.333526	0.201816	0.960133
3	XGBoost	0.957090	0.323693	0.194389	0.966777



Results

Case 1: Train (45k), val (10k) and test (10k) from labeled set only

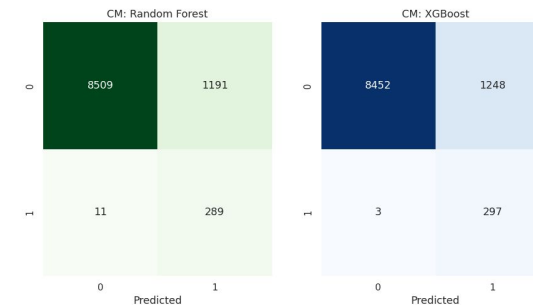
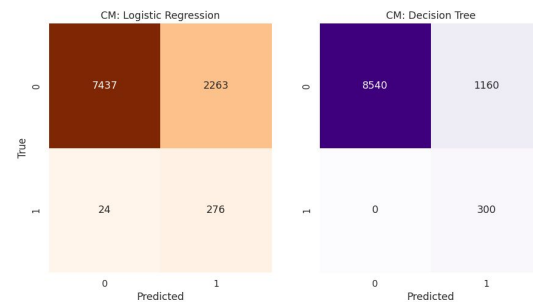
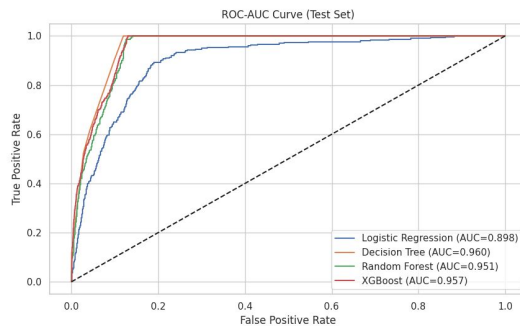
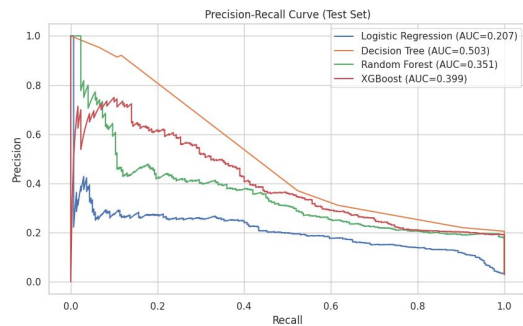
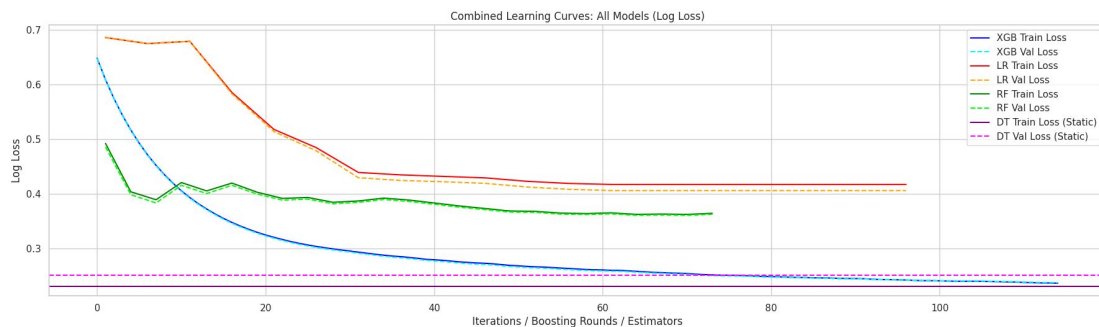
	Model	Opt_Threshold	Cost / Row	Total Cost (Test)	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.838510	148.49500	1484950.0	0.030100	1.000000	0.058441	0.505815
1	Decision Tree	0.346339	70.58840	705884.0	0.205781	0.993355	0.340935	0.940353
2	Random Forest	0.297031	104.79583	1047958.3	0.181486	0.990033	0.306742	0.945694
3	XGBoost	0.939554	173.04178	1730417.8	0.208924	0.980066	0.344425	0.964505



Results

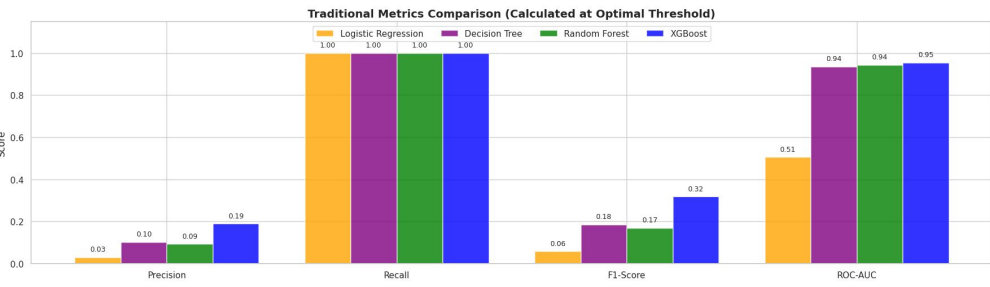
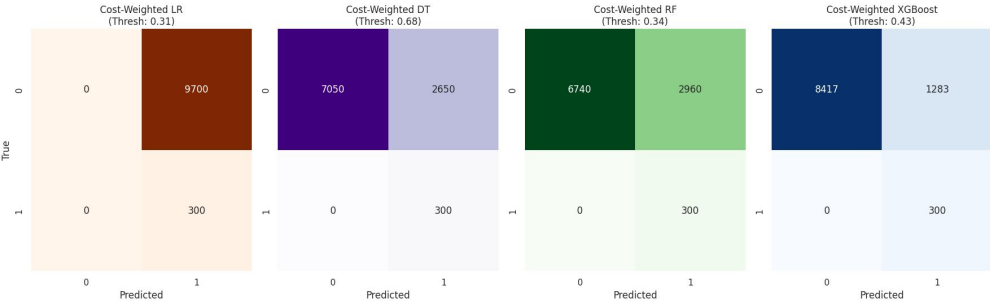
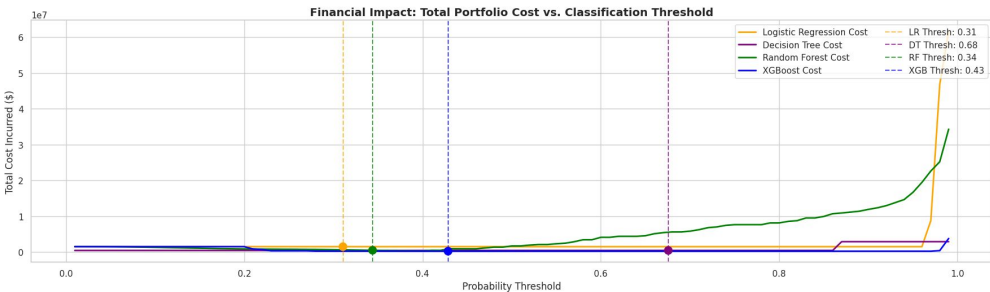
Case 2: Train (45k) and val (20k) from labeled, while Test (10k) from unlabeled set

	Model	ROC-AUC	F1-Score	Precision	Recall
0	Logistic Regression	0.897624	0.194435	0.108704	0.920000
1	Decision Tree	0.960261	0.340909	0.205479	1.000000
2	Random Forest	0.951370	0.324719	0.195270	0.963333
3	XGBoost	0.956589	0.321951	0.192233	0.990000



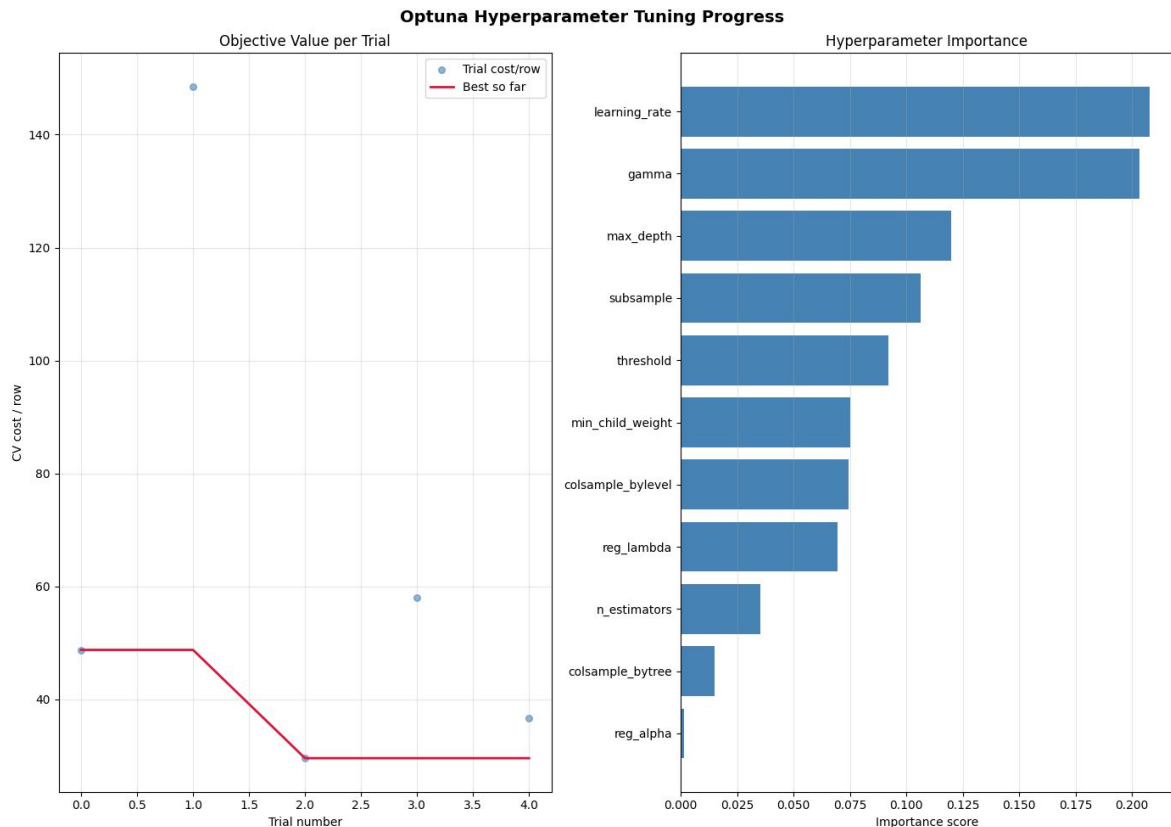
Results

Case 2: Train (45k) and val (20k) from labeled, while Test (10k) from unlabeled set



Grid Search (baseline) model

Case 1: Train (45k), val (10k) and test (10k)
from labeled set only



Final Test Set Results

Threshold	: 0.1085
Best boost round	: 237
Total business cost	: 225,250.00
Cost per row	: 22.5250
TP / FP / TN / FN	: 301 / 1301 / 8398 / 0
Precision	: 0.1879
Recall	: 1.0000
F1	: 0.3163

XGB params:

max_depth	= 5
min_child_weight	= 4
gamma	= 0.45606998421703593
learning_rate	= 0.12448918446337819
subsample	= 0.5998368910791798
colsample_bytree	= 0.708540663048167
colsample_bylevel	= 0.7554487413172255
reg_alpha	= 2.4298880728901692e-08
reg_lambda	= 0.0019275890163896973

Various Active Learning strategies

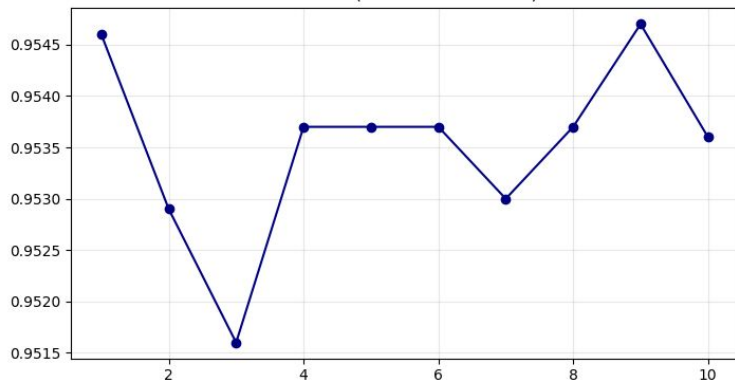
- **Cost-Weighted:** Selects highly uncertain predictions and scales them by their financial impact (insured amount) to prioritize reviewing costly borderline cases.
- **Uncertainty:** Selects the most ambiguous samples closest to the 50% decision boundary, where the model is least confident.
- **Diversity (Cluster + Uncertainty):** Selects a broad range of hard-to-predict samples by combining the model's uncertainty with spatial distances from K-means clusters to avoid querying redundant data points.
- **Expected Cost Reduction:** Selects samples with the highest potential financial regret by weighing the specific monetary penalties of making a false positive versus a false negative.

Active Learning (AL) Results

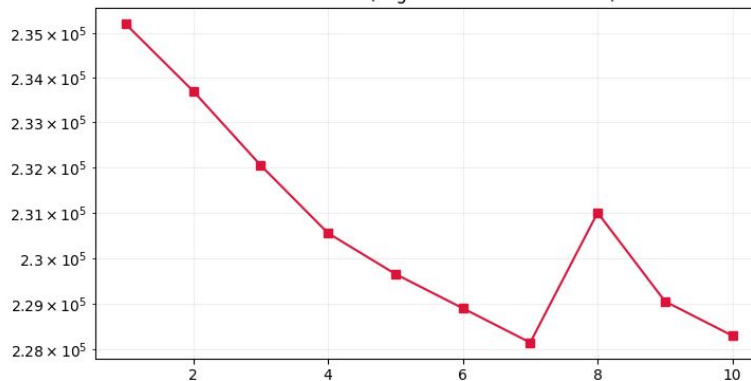
Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)

Cost-weighted uncertainty strategy

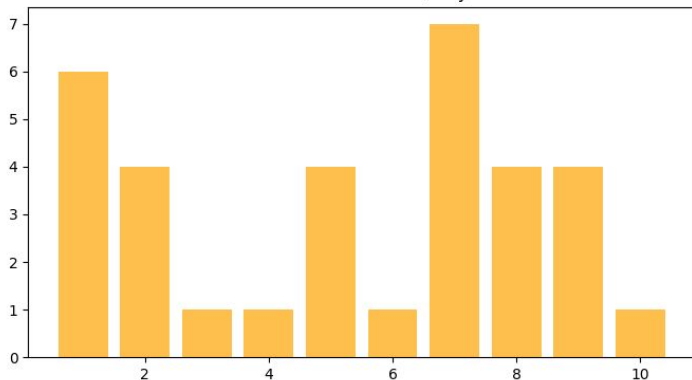
Test AUC (Model Performance)



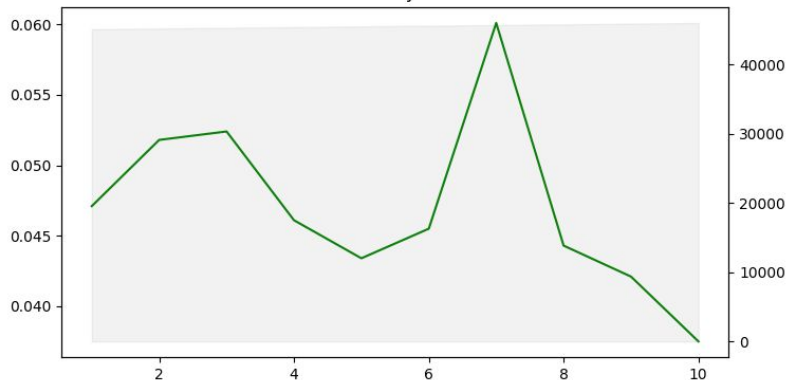
Business Cost (Log Scale - Lower is Better)



Frauds Found in Current Query Batch



Mean Uncertainty vs. Pool Size



Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)

Cost-weighted uncertainty strategy

Active Learning Progression Summary

loop	train_pool_size	unlabeled_remaining	fraud_in_query	mean_uncertainty	test_auc	test_business_cost
1	45100	18000	6	0.0471	0.9546	235200.0
2	45200	16000	4	0.0518	0.9529	233700.0
3	45300	14000	1	0.0524	0.9516	232050.0
4	45400	12000	1	0.0461	0.9537	230550.0
5	45500	10000	4	0.0434	0.9537	229650.0
6	45600	8000	1	0.0455	0.9537	228900.0
7	45700	6000	7	0.0601	0.9530	228150.0
8	45800	4000	4	0.0443	0.9537	231000.0
9	45900	2000	4	0.0421	0.9547	229050.0
10	46000	0	1	0.0375	0.9536	228300.0

BASELINE vs. FINAL ACTIVE LEARNING MODEL

Metric	Baseline	AL Final	Delta
AUC	0.9533	0.9536	+0.0003
Business Cost	\$232,950.00	\$228,300.00	\$-4,650.00
Fraud Found in AL	-	33	N/A

💡 Summary: AL reduced business cost by 2.00%

The model was enriched with 33 high-value fraud cases.

Non-Sequential (NSL) Results

Train from unlabeled set.

Test (10k) from unlabeled set

(To have consistent data for performance comparison between AL and NSL)

Non-Sequential Learning Progression Summary

train_pool_size	fraud_in_query	test_auc	test_business_cost
46000	27	0.9542	231750.0

Baseline → Final NSL model (train-set, indicative only)

AUC : 0.9542

AUC : 0.9533 → 0.9542 ($\Delta +0.0009$)

Business cost: 232,950 → 231,750 ($\Delta -1,200$)

AL vs NSL comparison

Train from unlabeled set.

Test (10k) from unlabeled set

(To have consistent data for performance comparison between AL and NSL)

Head-to-head comparison on final test set (10,000 rows)		
model	AL Model (sequential, final)	Non-Sequential Model (random 1k batch)
auc	0.9536	0.9542
biz_cost	228300.0000	231750.0000
cost_per_row	22.8300	23.1750
tp	300.0000	300.0000
fp	1322.0000	1345.0000
tn	8378.0000	8355.0000
fn	0.0000	0.0000
precision	0.1850	0.1824
recall	1.0000	1.0000
f1	0.3122	0.3085

Δ AL - Non-Sequential (positive = AL better for AUC/recall/f1, lower for cost)

AUC : -0.0006

Business cost: -3,450.00  AL cheaper

Recall : +0.0000

F1 : +0.0037

Baseline vs AL vs NSL comparison on Out-of-Sample (OOS) test

Head-to-head comparison on final held-out test set (28,072 rows)

model	Baseline (labeled pool only)	Non-Sequential (random 1k unlabeled)	Active Learning (sequential, 10 loops)
auc	0.9559	0.9567	0.9554
biz_cost	618750.0000	620400.0000	609900.0000
cost_per_row	22.0415	22.1003	21.7263
tp	843.0000	843.0000	843.0000
fp	3563.0000	3574.0000	3504.0000
tn	23666.0000	23655.0000	23725.0000
fn	0.0000	0.0000	0.0000
precision	0.1913	0.1909	0.1939
recall	1.0000	1.0000	1.0000
f1	0.3212	0.3205	0.3249

Δ AL vs Non-Sequential (positive = AL better for AUC/recall/f1)

AUC : -0.0013

Business cost: -10,500.00  cheaper

Recall : +0.0000

F1 : +0.0044

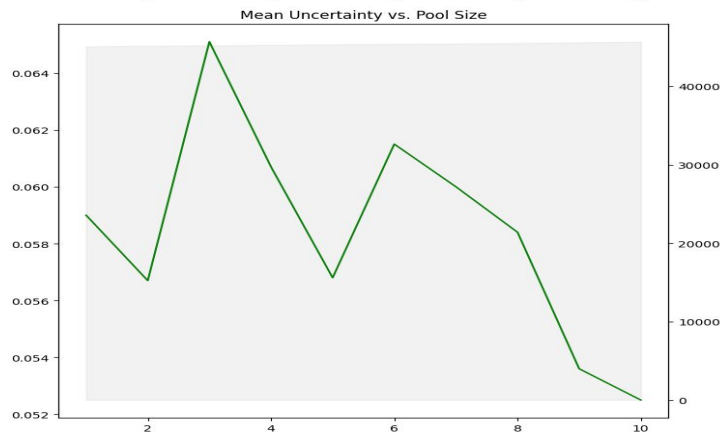
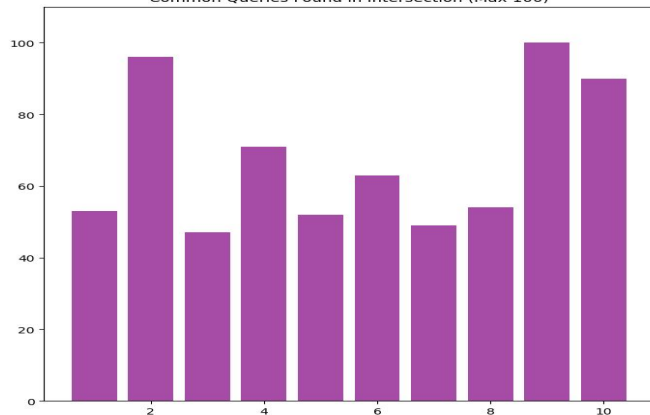
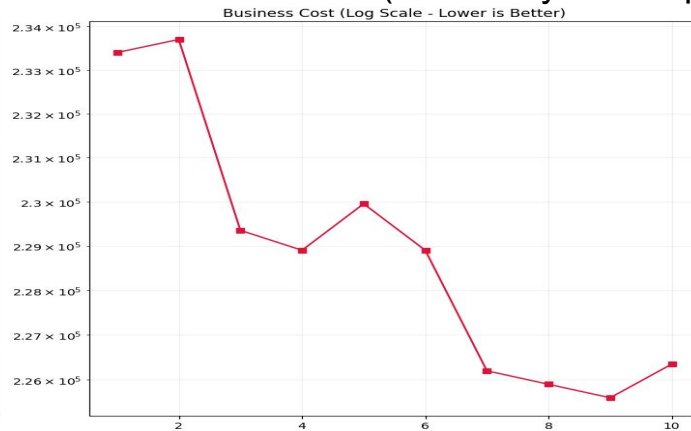
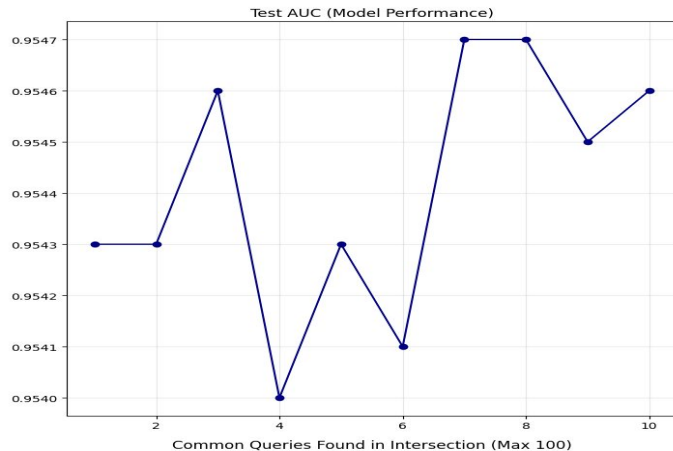
df_unlabeled total rows : 30,000

seen rows (union, deduped) : 1,948

final held-out test rows : 28,072

Hybrid Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see Hybrid AL progress)



Hybrid AL Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see Hybrid AL progress)

Hybrid Active Learning Progression Summary

loop	train_pool_size	unlabeled_remaining	common_queries	fraud_in_query	mean_uncertainty	test_auc	test_business_cost
1	45053	18000	53	0	0.0590	0.9543	233400.0
2	45149	16000	96	0	0.0567	0.9543	233700.0
3	45196	14000	47	0	0.0651	0.9546	229350.0
4	45267	12000	71	0	0.0607	0.9540	228900.0
5	45319	10000	52	0	0.0568	0.9543	229950.0
6	45382	8000	63	0	0.0615	0.9541	228900.0
7	45431	6000	49	0	0.0600	0.9547	226200.0
8	45485	4000	54	0	0.0584	0.9547	225900.0
9	45585	2000	100	0	0.0536	0.9545	225600.0
10	45675	0	90	0	0.0525	0.9546	226350.0

BASELINE vs. FINAL HYBRID ACTIVE LEARNING MODEL

Metric	Baseline	HYBRID AL	Delta
AUC	0.9533	0.9546	+0.0013
Business Cost	\$232,950.00	\$226,350.00	\$-6,600.00
Total Queries Made	-	675	N/A
Fraud Found in AL	-	0	N/A

💡 Summary: Hybrid AL reduced business cost by 2.83%

The model queried 675 highly robust rows, yielding 0 high-value fraud cases.

Baseline vs AL vs NSL vs Hybrid AL comparison on Out-of-Sample (OOS) test

Head-to-head comparison on final held-out test set (27,707 rows)

model	Baseline (labeled pool only)	Non-Sequential (random 1k unlabeled)	Active Learning (Cost-weighted Uncertainty)	Hybrid AL (Multi-Strategy)
auc	0.9553	0.9561	0.9547	0.9563
biz_cost	617400.0000	616800.0000	609300.0000	607950.0000
cost_per_row	22.2832	22.2615	21.9908	21.9421
tp	843.0000	843.0000	843.0000	843.0000
fp	3554.0000	3550.0000	3500.0000	3491.0000
tn	23310.0000	23314.0000	23364.0000	23373.0000
fn	0.0000	0.0000	0.0000	0.0000
precision	0.1917	0.1919	0.1941	0.1945
recall	1.0000	1.0000	1.0000	1.0000
f1	0.3218	0.3220	0.3251	0.3257

HYBRID AL vs AL (Cost-Weighted Uncertainty)

Δ Hybrid AL vs AL (positive = Hybrid AL better for AUC/recall/f1)

AUC : +0.0016

Business cost: -1,350.00  cheaper

Recall : +0.0000

F1 : +0.0006

df_unlabeled total rows : 30,000

seen rows (union, deduped) : 2,313

final held-out test rows : 27,707

AL (various strategies) vs Hybrid AL comparison on Out-of-Sample (OOS) test

Head-to-head comparison on final held-out test set (27,583 rows)

		AL (Simple)	AL (Cost-Weighted)	AL (Diversity)	AL (Expected Cost)	Hybrid AL
auc		0.9563	0.9558	0.9572	0.9567	0.9570
biz_cost		589200.0000	589350.0000	587100.0000	588600.0000	587250.0000
df_unlabeled total rows	: 30,000					
cost_per_row		21.3610	21.3664	21.2848	21.3392	21.2903
seen rows (union, deduped)	: 2,417					
tp		840.0000	840.0000	840.0000	840.0000	840.0000
final held-out test rows	: 27,583					
fp		3368.0000	3369.0000	3354.0000	3364.0000	3355.0000
tn		23375.0000	23374.0000	23389.0000	23379.0000	23388.0000
fn		0.0000	0.0000	0.0000	0.0000	0.0000
precision		0.1996	0.1996	0.2003	0.1998	0.2002
recall		1.0000	1.0000	1.0000	1.0000	1.0000
f1		0.3328	0.3327	0.3337	0.3331	0.3337

Δ Hybrid AL vs AL (Simple) (positive = Hybrid AL better for AUC/recall/f1)

AUC : +0.0007

Business cost: -1,950.00  cheaper

Recall : +0.0000

F1 : +0.0009

Δ Hybrid AL vs AL (Cost-Weighted) (positive = Hybrid AL better for AUC/recall/f1)

AUC : +0.0012

Business cost: -2,100.00  cheaper

Recall : +0.0000

F1 : +0.0010

Δ Hybrid AL vs AL (Expected Cost) (positive = Hybrid AL better for AUC/recall/f1)

AUC : +0.0003

Business cost: -1,350.00  cheaper

Recall : +0.0000

F1 : +0.0006

Δ Hybrid AL vs AL (Diversity) (positive = Hybrid AL better for AUC/recall/f1)

AUC : -0.0002

Business cost: +150.00  more expensive

Recall : +0.0000

F1 : +0.0000

References

A] UK Data Sources Used

1. **Population, Occupation, Housing:**
 - 1.1. ONS Labour Force Survey
 - 1.2. UK Housing Survey 2023
 - 1.3. Gov.uk English Housing Data
2. **Risk Maps:**
 - 2.1. Environment Agency Flood Risk Map
 - 2.2. Met Office Severe Weather Patterns
3. **Insurance Market:**
 - 3.1. Association of British Insurers (ABI) Home Insurance Report
 - 3.2. Insurance Fraud Bureau (IFB) Fraud Typology
 - 3.3. FCA Household Insurance Market Analytics
4. **Payment Behaviour:**
 - 4.1. Experian UK Financial Wellness Index
 - 4.2. FCA Consumer Risk Survey

C] Regulatory & Government Sources

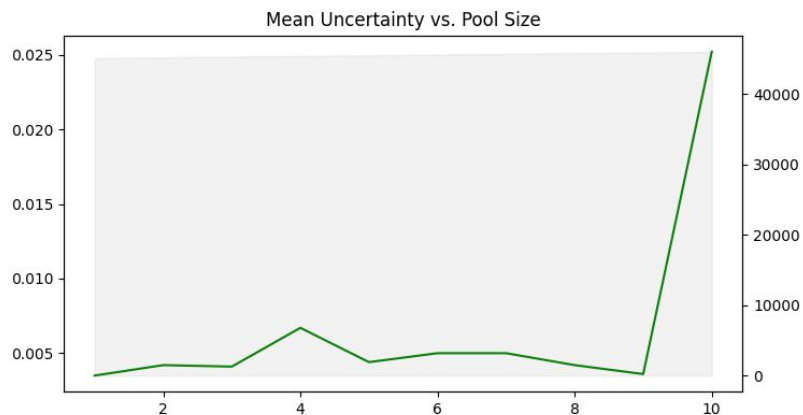
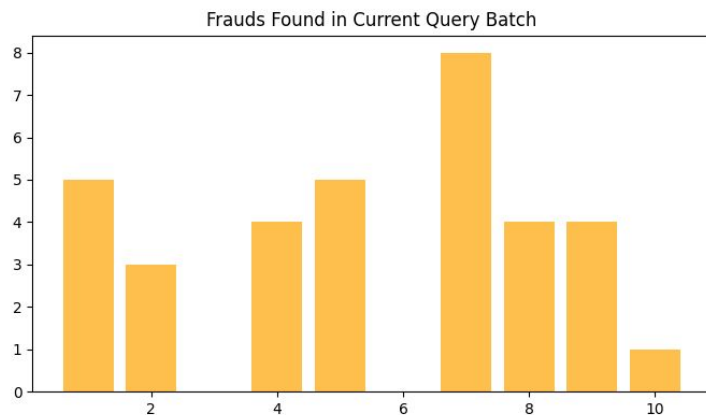
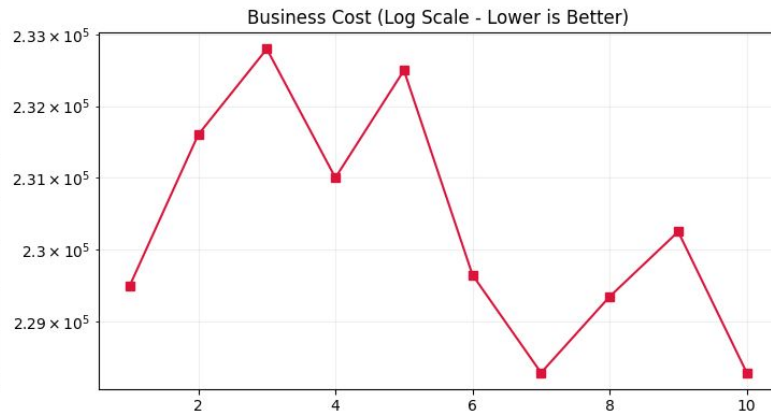
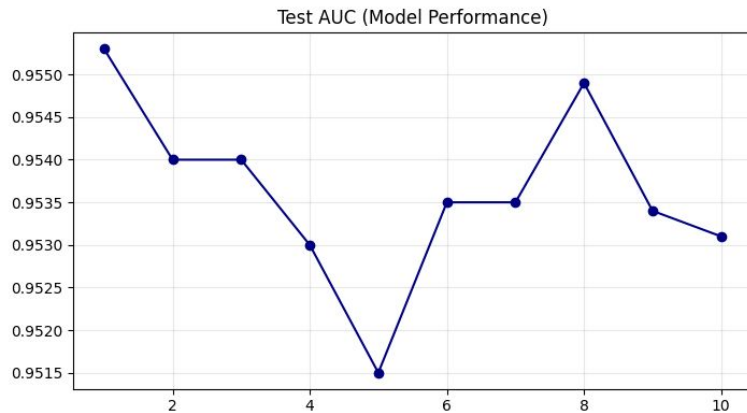
1. **Financial Conduct Authority (FCA)** (2024). *Home Insurance Pricing Practices*.
 - 1.1. Auto-renewal regulations, price walking reforms
2. **Environment Agency** (2024). *Flood Risk Assessment*.
 - 2.1. 10% of English homes in flood zones

B] Academic & Industry Sources

1. **Mordor Intelligence** (2024). *UK Home Insurance Market Report*.
<https://www.mordorintelligence.com/industry-reports/uk-home-insurance-market>
 - 1.1. Property type distribution, premium ranges
2. **Mintel** (2024). *UK Home Insurance Market Report 2024*.
<https://store.mintel.com/report/uk-home-insurance-market-report-2024>
 - 2.1. Market size (£9.4B forecast), ownership rates
3. **GlobalData** (2024). *UK Household Insurance Market Dynamics*.
 - 3.1. GWP growth, premium inflation, claims landscape
4. **Confused.com** (2024). *UK Home Insurance Statistics*.
<https://www.confused.com/home-insurance/home-insurance-statistics>
 - 4.1. Average premiums (£396), claim frequencies
5. **Uswitch** (2025). *Home Insurance Statistics*.
<https://www.uswitch.com/home-insurance/studies/home-insurance-statistics-uk/>
 - 5.1. Weather claims (£585M), regional variations
6. **Association of British Insurers (ABI)** (2024). *Insurance Fraud Statistics*.
 - 6.1. 10-15% fraud rate benchmark
7. **Office for National Statistics (ONS)** (2021). *UK Census - Housing Stock*.
 - 7.1. Property age distribution, urban/rural split
8. **English Housing Survey** (2023). *Housing Stock Report*.
 - 8.1. Property condition, age bands
9. [Economic and social cost of fraud 2023 to 2024](#)
10. [Uncovering the real cost of fraud](#)
11. [Insurance Fraud Taskforce: interim report](#)

Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)
Uncertainty strategy



Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)
Uncertainty strategy

Active Learning Progression Summary						
loop	train_pool_size	unlabeled_remaining	fraud_in_query	mean_uncertainty	test_auc	test_business_cost
1	45100	18000	5	0.0035	0.9553	229500.0
2	45200	16000	3	0.0042	0.9540	231600.0
3	45300	14000	0	0.0041	0.9540	232800.0
4	45400	12000	4	0.0067	0.9530	231000.0
5	45500	10000	5	0.0044	0.9515	232500.0
6	45600	8000	0	0.0050	0.9535	229650.0
7	45700	6000	8	0.0050	0.9535	228300.0
8	45800	4000	4	0.0042	0.9549	229350.0
9	45900	2000	4	0.0036	0.9534	230250.0
10	46000	0	1	0.0252	0.9531	228300.0

BASELINE vs. FINAL ACTIVE LEARNING MODEL			
Metric	Baseline	AL Final	Delta
AUC	0.9533	0.9531	-0.0002
Business Cost	\$232,950.00	\$228,300.00	\$-4,650.00
Fraud Found in AL	-	34	N/A

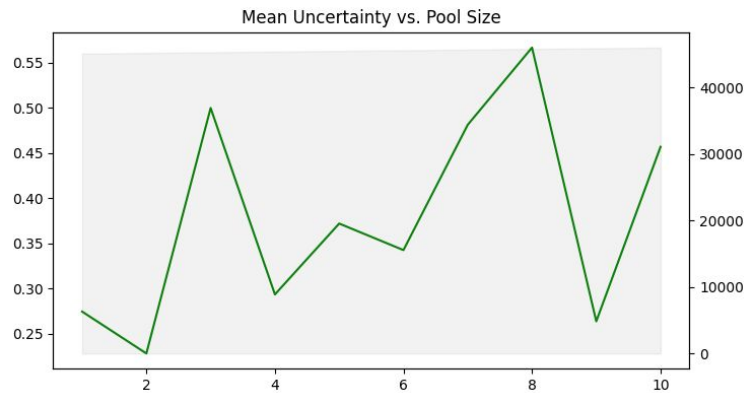
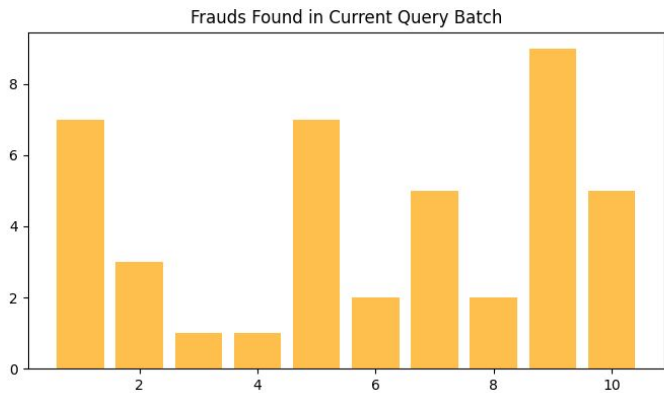
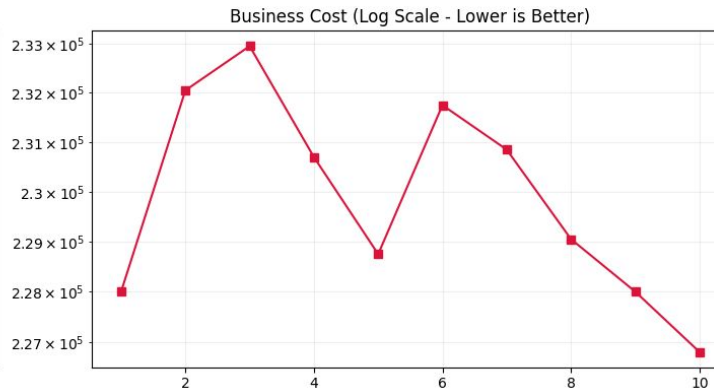
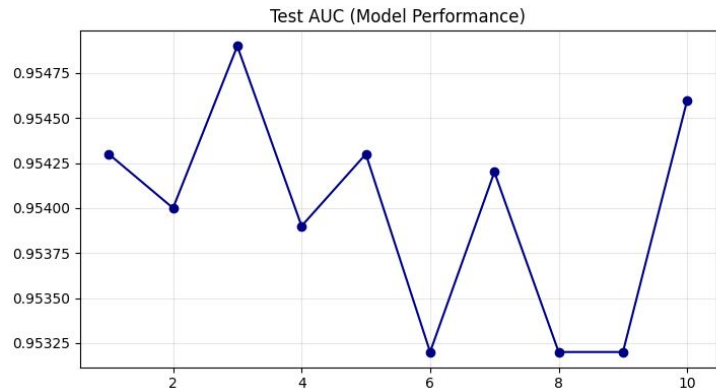
💡 Summary: AL reduced business cost by 2.00%
The model was enriched with 34 high-value fraud cases.

Active Learning (AL) Results

Train from unlabeled set.

Test (10k) from unlabeled set
(To see AL progress)

**Diversity (K-Means Distance-Weighted
Uncertainty) strategy**



Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)

**Diversity (K-Means Distance-Weighted
Uncertainty) strategy**

Active Learning Progression Summary

loop	train_pool_size	unlabeled_remaining	fraud_in_query	mean_uncertainty	test_auc	test_business_cost
1	45100	18000	7	0.2744	0.9543	228000.0
2	45200	16000	3	0.2280	0.9540	232050.0
3	45300	14000	1	0.4999	0.9549	232950.0
4	45400	12000	1	0.2934	0.9539	230700.0
5	45500	10000	7	0.3719	0.9543	228750.0
6	45600	8000	2	0.3425	0.9532	231750.0
7	45700	6000	5	0.4813	0.9542	230850.0
8	45800	4000	2	0.5668	0.9532	229050.0
9	45900	2000	9	0.2636	0.9532	228000.0
10	46000	0	5	0.4569	0.9546	226800.0

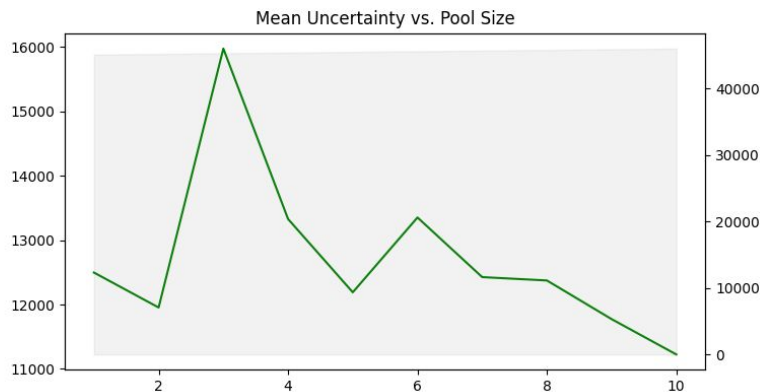
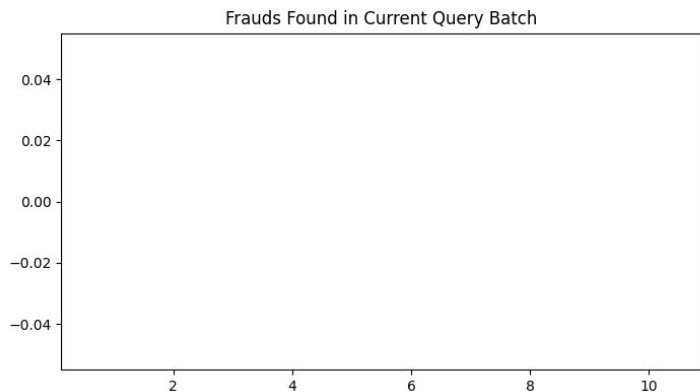
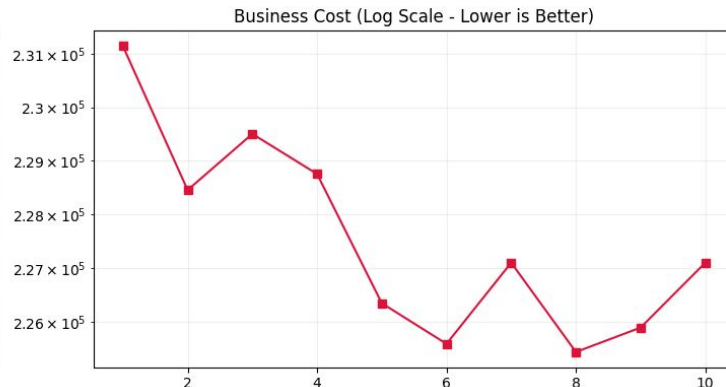
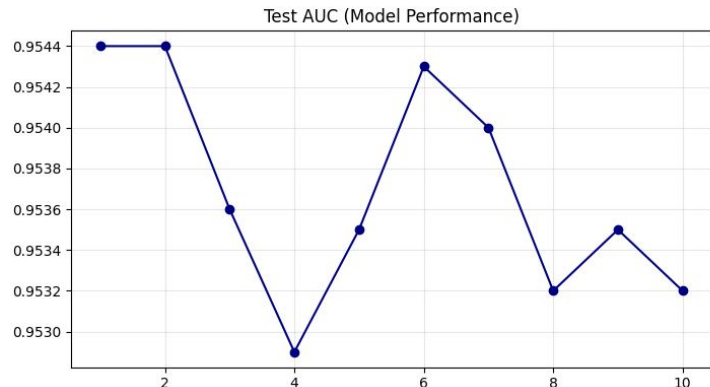
BASELINE vs. FINAL ACTIVE LEARNING MODEL

Metric	Baseline	AL Final	Delta
AUC	0.9533	0.9546	+0.0013
Business Cost	\$232,950.00	\$226,800.00	\$-6,150.00
Fraud Found in AL	-	42	N/A

💡 Summary: AL reduced business cost by 2.64%
The model was enriched with 42 high-value fraud cases.

Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)
Expected Cost Reduction strategy



Active Learning (AL) Results

Train from unlabeled set.
Test (10k) from unlabeled set
(To see AL progress)
Expected Cost Reduction strategy

Active Learning Progression Summary

loop	train_pool_size	unlabeled_remaining	fraud_in_query	mean_uncertainty	test_auc	test_business_cost
1	45100	18000	0	12497.1283	0.9544	231150.0
2	45200	16000	0	11954.1163	0.9544	228450.0
3	45300	14000	0	15974.5543	0.9536	229500.0
4	45400	12000	0	13328.3489	0.9529	228750.0
5	45500	10000	0	12190.3850	0.9535	226350.0
6	45600	8000	0	13352.0141	0.9543	225600.0
7	45700	6000	0	12426.2112	0.9540	227100.0
8	45800	4000	0	12373.9245	0.9532	225450.0
9	45900	2000	0	11773.7869	0.9535	225900.0
10	46000	0	0	11225.0611	0.9532	227100.0

BASELINE vs. FINAL ACTIVE LEARNING MODEL

Metric	Baseline	AL Final	Delta
AUC	0.9533	0.9532	-0.0001
Business Cost	\$232,950.00	\$227,100.00	\$-5,850.00
Fraud Found in AL	-	0	N/A

💡 Summary: AL reduced business cost by 2.51%
The model was enriched with 0 high-value fraud cases.

THANK YOU

